

ICMP Protocol (Internet Control Message Protocol)

ICMP (Internet Control Message Protocol) is a network layer protocol used to check network connectivity and report network errors. It does not transfer user data like TCP or UDP. Common ICMP messages include *Echo Request* and *Echo Reply*, which are generated using the `ping` command. Echo Request is sent to check whether a device is reachable, and Echo Reply is sent back by the destination device to confirm that it is reachable.

- ICMP messages generated by the Ping program;
 - ICMP messages generated by the Traceroute program;
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- ❖ Open Command Prompt (CMD) and Wireshark. Start packet capture in Wireshark.
 - ❖ Run the command `ping -n 10 www.sandesh.com`.
 - ❖ After all 10 ping requests and replies are completed, stop the capture.
 - ❖ Apply the `icmp` filter and use the same captured packets to answer all the ICMP (Ping) practical questions.

1.What is the IP address of your host? What is the IP address of the destination host?

2.Why doesn't an ICMP packet have source and destination port numbers?

ICMP does not have source and destination port numbers because it is a network layer protocol for sending error messages and checking network connectivity, not a transport layer protocol like TCP or UDP. It uses only IP addresses to identify the source and destination.

3. Examine one of the ping request packets sent by your host. What are the ICMP type and code numbers? What other fields does this ICMP packet have? How many bytes are the checksum, sequence number and identifier fields?

Answer:

- **ICMP Type = 8 (Echo Request)**
- **ICMP Code = 0**

- Type
- Code
- Checksum
- Identifier
- Sequence Number

Checksum = 2 bytes

Identifier = 2 bytes

Sequence Number = 2 bytes

4. What is the IP address of your host? What is the IP address of the target destination host? (using tracert command)

Traceroute

Tracing route to www.sandesh.com [172.66.163.38]

Here, DNS resolved www.sandesh.com to:

172.66.163.38

Ping

Pinging www.sandesh.com [104.20.44.68]

Here, DNS resolved the **same website** to:

104.20.44.68

5) If ICMP uses UDP packets instead, will the Protocol Number still be 1? If not, what will it be?

-> No. The protocol number will be 17 because UDP uses Protocol Number 17, while ICMP uses Protocol Number 1.

6) Look at the last three ICMP packets received by your computer. How are they different from the ICMP Error packets? Why are they different?

The last three ICMP packets are different because they are sent by the destination host, not by intermediate routers. They are normal reply packets, whereas the ICMP Error packets are sent by routers before the packet reaches the destination.

7) What is the ICMP Type and Code for Echo Reply

8) What is the purpose of the tracer command? What is the purpose of the ping command?